University of California, Berkeley - College of Engineering, EECS Dept

Summer 2026 Instructor: Della Dastur 2026-07-02

CS10 Quiz 1

Last Name	
First Name	
Student ID Number	
Name of the person to your Left	
Name of the person to your Right	
All my work is my own. I had no prior knowledge of the exam contents nor will I share the contents with others in CS10 who have not taken it yet. ***(please sign)***	

Instructions

- This booklet contains 4 double-sided pages including this cover page. Put all answers on these pages; don't hand in any stray pieces of paper.
- Please put all smart watches, cell phones, and beepers at the back of the room. Remove all hats, headphones, and watches.
- You have 180 minutes to complete this exam. This final is a closed book, no computers, watches, no PDAs, no cell phones, no calculators. There may be partial credit for incomplete answers; write as much of the solution as you can. When we provide a blank, please fit your answer within the space provided.

Points

Abstraction	Binary - Hex - Dec	Iteration	Domain & Range	Booleans	Functions	HOFs
2	4	6	6	6	4	12

Question 1 - Abstraction *(2 Points)*

How does moving from **BEFORE** to **AFTER** affect *Abstraction* and its related concepts?

PART A

BEFORE	AFTER
	

Detail Removal: ◯ decreases ◯ increases ◯ no change

Generalization: ◯ decreases ◯ increases ◯ no change

Abstraction: ◯ decreases ◯ increases ◯ no change

PART B

Specification for 

BEFORE	AFTER
Inputs: a number	**Inputs:** a number
Output: Double the input	**Output:** Double the input
Example: 	**Example:** 
	How we implemented it: We called the input parameter `n`. We took `n`, then added it to itself and set a temporary variable `answer` to that sum. Then we returned that `answer` variable (which was effectively `n + n`).

Detail Removal: ◯ decreases ◯ increases ◯ no change

Generalization: ◯ decreases ◯ increases ◯ no change

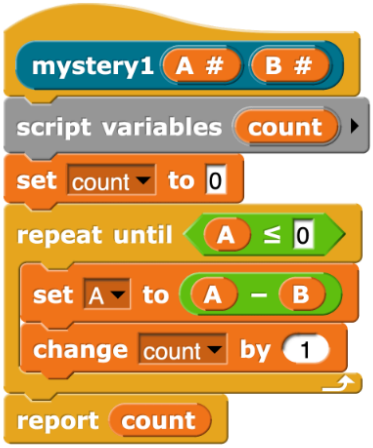
Abstraction: ◯ decreases ◯ increases ◯ no change

Question 2 - Number Representation *(4 Points)*

What is $1010_2 + 13_{10}$?

- ○ 25_{16}
- ○ $1F_{16}$
- ○ 17_{16}
- ○ $1D_{16}$
- ○ 21_{16}
- ○ $1B_{16}$
- ○ 19_{16}
- ○ 23_{16}

Question 3 - Iteration *(6 points)*
Part A



a) Given the code above, what does  report when **A** is set to 12, and **B** is set to 3? (Fill in the blank)

Answer: _____

Part B

b) What does `mystery2` compute if `A` and `B` are positive integers?

- ○ `A` - `B`
- ○ `B` - `A`
- ○ The minimum value of `A` and `B`
- ○ The maximum value of `A` and `B`
- ○ The number of times `B` can be added up before it is bigger than `A`
- ○ The number of times `A` can be added up before it is bigger than `B`
- ○ The remainder when `A` is divided by `B`
- ○ The remainder when `B` is divided by `A`
- ○ None of the Above

Question 4 - Domain and Range *(6 points)*

If `outer Function inner value` is error-free, with the following properties:

- `inner` **range:** word
- `outer` **range:** list
- `inner` **domain:** Boolean
- `outer` **domain:** character

...and we want `new Function input` to be error-free, which of the following types would be allowed? (Don't forget about subtypes! For example, if *numbers* were allowed, then a *digit* would **also** be allowed, since a *digit* is also a *number*.) `new` **range**:

☐ Boolean ☐ list ☐ sentence ☐ word ☐ character ☐ letter

`new` **domain**: ☐ Boolean ☐ list ☐ sentence ☐ word ☐ character ☐ letter

`input`: ☐ Boolean ☐ list ☐ sentence ☐ word ☐ character ☐ letter

Question 5 - Conditional Operators and Booleans *(6 points)*

Part 1

Suppose the output of `Pred` is 

.

What can you say for sure about **A**?

- ○ **A** must be true
- ○ **A** must be false
- ○ Nothing

What can you say for sure about **B**?

- ○ **B** must be true
- ○ **B** must be false
- ○ Nothing

Part 2

Suppose the output of `Pred` is 

.

What can you say for sure about **A**?

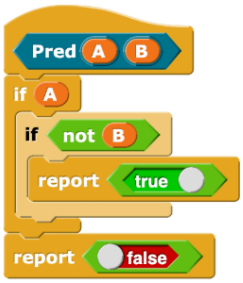
- ○ **A** must be true
- ○ **A** must be false
- ○ Nothing

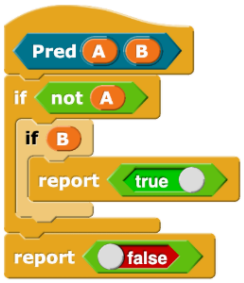
What can you say for sure about **B**?

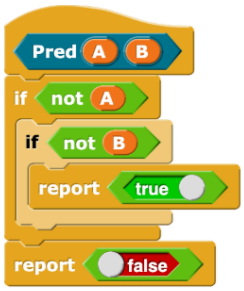
- ○ **B** must be true
- ○ **B** must be false
- ○ Nothing

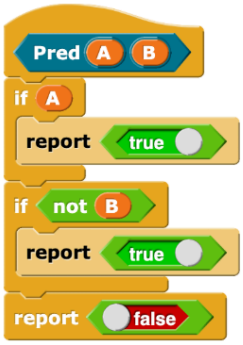
Part 3

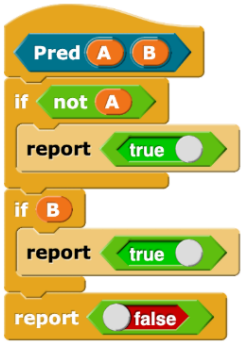
Which definition below computes the same logical function as `Pred` at the top?

○ 

○ 

○ 

○ 

○ 

○ 

Question 6 - Functions (4 points)

If we were given three functions:

$$R(x) = x + 4$$

$$S(x) = x - 3$$

$$K(x) = x \times 2$$

... and you want to calculate:

$$(x - 3 + 4) \times 2$$

... how would you compose the three functions to get that?

- ○ $R(K(S(x)))$
- ○ $S(R(K(x)))$
- ○ $R(S(K(x)))$
- ○ $K(S(R(x)))$
- ○ $S(K(R(x)))$
- ○ $K(R(S(x)))$

Question 7 - HOFs *(12 points)*

We want to take the command blocks `delete`, `replace`, and `add` and convert them from commands to reporters so that they return *new* lists and don't modify their input list. What would be needed to reimplement them as reporters as shown in each example? (we also add a bonus reporter, called `only the ith elements`).

a)

- ○ Can be done with 1 **map**
- ○ Can be done with 1 **keep**
- ○ Must be done with a **map** and **keep** together
- ○ None of these

b)

- ○ Can be done with 1 **map**
- ○ Can be done with 1 **keep**
- ○ Must be done with a **map** and **keep** together
- ○ None of these

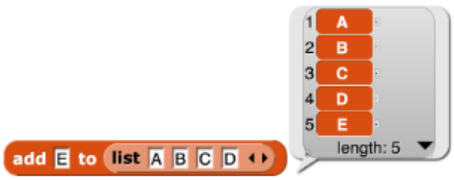
c)

- ○ Can be done with 1 **map**
- ○ Can be done with 1 **keep**
- ○ Must be done with a **map** and **keep** together
- ○ None of these

d)

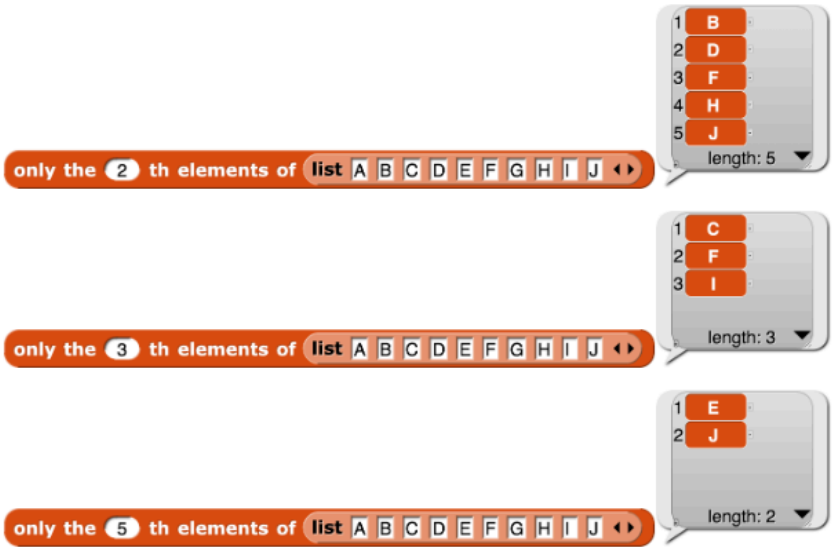
- ○ Can be done with 1 **map**
- ○ Can be done with 1 **keep**
- ○ Must be done with a **map** and **keep** together
- ○ None of these

e)



- ○ Can be done with 1 **map**
- ○ Can be done with 1 **keep**
- ○ Must be done with a **map** and **keep** together
- ○ None of these

f)



- ○ Can be done with 1 **map**
- ○ Can be done with 1 **keep**
- ○ Must be done with a **map** and **keep** together
- ○ None of these